A Comparative Analysis of Supervised Learning Models for Heart Disease Prediction

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A. Abstract

This study compares five supervised machine learning models for predicting heart disease based on various health and lifestyle factors, using a subset of the well-known UCI heart disease dataset. The project employs a uniform preprocessing pipeline for all models, including handling missing values, scaling, outlier detection, and categorical encoding, as well as hyperparameter tuning for training models using GridSearchCV and Optuna. Overall, all models performed reasonably well, with Logistic Regression achieving the best performance in heart disease prediction, followed closely by KNN, while the other models were slightly less performant, but overall comparable. This is likely due to the fact that the data is already well structured, as well as the fact that the dataset is of modest size. At an F1 score of 87.9%, the model shows promise in heart disease prediction with reliable accuracy.

All in all, this study reiterates the benefits of machine learning for medical research, and in particular, early heart disease prediction, which can inform early prevention and treatment plans, as well as avoiding more critical, avoidable complications.

B. Introduction

Cardiovascular diseases (CVDs) are the number one cause of death globally, taking an estimated 17.5 million lives each year, which accounts for 31% of all deaths worldwide (El-Sofany et al., 2024). Four out of five CVD deaths are due to heart attacks and strokes, and one-third of these deaths occur prematurely in people under 70 years of age. Heart failure is a common event caused by CVDs.

As CVD impacts a large proportion of the population in many ways, this project aims to leverage available data to improve early heart disease prediction using machine learning. It poses the question of which machine learning model predicts heart disease the best given our dataset and engineered features. Current research includes support vector machines, XGBoost, bagging, decision trees, and random forests on UCI heart disease and Cleveland heart disease datasets, as well as other real-world and private datasets. XGBoost is suggested to be the most high-performing algorithm (El-Sofany et al., 2024). Overall, machine learning has proved to be promising, with high accuracy achieved in heart disease prediction.

This particular dataset is a subset of the Heart Disease dataset from UC Irvine. The original contains 76 attributes, but all published experiments refer to using a subset of only fourteen of them, with the Cleveland dataset being the only used in ML research to date (Janosi et al., 1988).

In this comparison study, five machine learning models are selected: KNN, Random Forest, Neural Network, Logistic Regression, and XGBoost. The choice of XGBoost as a novel model is informed by current literature, and hypothesised to be the most well-performed model in this study. The choice of Random Forest and Neural Networks stems from the findings that these two models generally receive less attention in heart disease research compared to decision trees, logistic regression, and Naive Bayes, despite high accuracy in certain contexts, and the fact that these are strong models generally, especially given the tabular structure of the data (Mao et al., 2025).

C. Data Collection, Preprocessing, and Representation

```
Summary statistics for numerical variables
          Age  RestingBP   MaxHR  Oldpeak  Cholesterol
min     28.00       0.00   60.00    -2.60         0.00
median  54.00     130.00  138.00     0.60       223.00
max     77.00     200.00  202.00     6.20       603.00
std      9.43      18.51   25.46     1.07       109.38
```

Table 1. Summary statistics of numerical variables in Heart Disease dataset

```
Average age for male versus female participants
      Age
Sex
F    52.0
M    54.0

Label probability of men with heart disease: 0.63
```

```
Average age for male versus female participants with heart disease
      Age
Sex
F    56.0
M    56.0

Label probability of women with heart disease: 0.26
```

Table 2. Summary of demographic-based heart-disease statistics

```
Data variance pre-NA-handling
     RestingBP  Cholesterol
std       18.0        59.15

Data variance post-NA-handling
     RestingBP  Cholesterol
std       18.0        53.43
```

```
Data variance pre-processing
     RestingBP  Cholesterol
std       18.0        59.15

Data variance post-NA and outlier-handling
     RestingBP               Cholesterol
std       0.13                      0.21
```

Table 3. Impact of missing value handling prior to and after data proprocessing

Exploratory data analysis shows that the data mostly came from middle-aged male (Figure 1.5, Figure 2). The split in the label of the dataset is fairly balanced, as seen in Figure 3. It appears that male individuals have a higher probability of heart disease at 63% compared to female individuals at 26%. The average age of participants with heart disease is 56 for both males and females.

Data preprocessing was consistent for all models. In terms of missing data, entries with value 0 in the numerical columns were treated as missing values and encoded as NA for analysis. For RestingBP, since there is only one missing value, the entry was dropped entirely. For Cholesterol, since 18% of the data was missing, imputation with the median of the training set on both the training and test set was the chosen approach. KNN imputation with k=5 neighbours was also experimented with; the effect on training results was minimal, if not less performant than median imputation, hence not implemented in the final version (See *Table 5* for these results). For outliers in RestingBP and Cholesterol, a logarithmic transformation approach was chosen to maintain the integrity of the data, as the newly imputed data is slightly skewed. The interpretation of the variance changes at this point and hence is no longer meaningful, but is noted for completeness. As KNN and in a way, logistic regression, benefit from scaling, this was done for the numerical features Age, RestingBP, MaxHR, Oldpeak, and Cholesterol. The StandardScaler was used to standardize these variables. Categorical variables were also encoded. Binary variables Sex, ExerciseAngina, AND FastingBS were binary-encoded; ChestPainType, RestingECG, and ST_Slope were one-hot encoded to not create ordinal orders that did not already exist in the data.

D. Model Comparison Study

KNN is implemented as our primary instance-based or non-generalizing learning benchmark. Based on documentation from scikit-learn, it does not attempt to construct a general internal model, but simply stores instances of the training data and computes the classification from a simple majority vote of the nearest neighbors of each point. Additionally, to find the prediction, it is calculated by averaging the outcomes with the local neighbourhood.

A random forest is a meta estimator to evaluate the effectiveness of ensemble learning on our clinical dataset. Based on documentation from scikit-learn, a random forest fits a number of decision tree classifiers on various sub-samples of the dataset and uses averaging to improve the predictive accuracy and control overfitting.

Neural Network represents a Deep Learning approach in our comparison study. Based on Universal Approximation Theorem, a feedforward neural network with a single hidden layer containing a finite number of neurons can approximate any continuous function on compact subsets of $\mathbb{R}^n$. In this study, a multi-layer perceptron is utilized with a non-linear ReLU activation function.

Logistic Regression is known for using categorical variables as a response variable. The logistic regression equation is given by $ln(\frac{\pi(x)}{1-\pi(x)}) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$ where $\pi(x) = \frac{e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}{1+e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}$ is the probability of heart disease and p is the number of features, modelling the relationship between the clinical features and the probability of heart disease using a linear predictor.

XGBoost was selected as the representative for boosting ensembles. According to GeeksforGeeks, the library implements an optimized version of gradient boosting where decision trees are trained sequentially. In this setup, each new tree is designed specifically to minimize the residual errors of the preceding learners, ultimately resulting in a high-performance ensemble model. First, the first model decision tree is trained on the data and documents the average prediction of the target variable. The algorithm then optimizes the model by minimizing the objective function: $E(\theta) = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k)$, where $l(y_i, \hat{y}_i)$ is the loss function that measures the difference between the true value y_i and the prediction $\hat{y}_i$ and $\Omega(f_k)$ is the regularization that penalizes complex trees to improve generalization. Next, the algorithm continues with the next tree, calculates the predictions using: $\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f_t(x_i)$, where $\hat{y}_i^{(t-1)}$ is the prediction from the previous iteration and $f_t(x_i)$ is the prediction of the tth tree for the ith data point, and repeats until a stopping criterion is met. Finally, it combines the predictions using this formula: by using this formula: $\hat{y}_i = \sum_{k=1}^{K} f_k(x_i)$, where $\hat{y}_i$ is the final predicted value for the ith data point, K is the total number of trees in the ensemble, and $f_k(x_i)$ represents the prediction of the kth tree for the ith data point.

Hyperparameter tuning was performed to ensure a fair comparison across models. Since these parameters are not learned from the data, they can strongly affect model complexity and performance.

For most models, we used GridSearchCV with 5-fold cross-validation to test different parameter combinations. The F1-score was used as the evaluation metric because it balances precision and recall, which is important in this task where missing a heart disease case can be more serious than a false positive.

For KNN, we tested the number of neighbours over [3, 5, 7, 9] and compared uniform and distance-based weighting. For Random Forest, parameters such as n_estimators, max_depth, and min_samples_leaf were tuned to control tree complexity and reduce overfitting. For Logistic Regression, the inverse of

regularization strength C was tested over a logarithmic range [10^{-3}, 10^{3}]. We also explored different solvers, namely lbfgs and liblinear from scikit-learn and different types of regularisation (L1 - Lasso vs L2 - Ridge).

For XGBoost, we used a sequential tuning approach to reduce computation time. In an initial round of tuning, key parameters such as max_depth, learning_rate, subsample, and colsample_bytree were selected to form a baseline. These values were fixed, and a second round of tuning was performed over min_child_weight, gamma, reg_alpha, and reg_lambda.

For the Neural Network, Optuna was used instead of GridSearchCV because of the large number of hyperparameters. Optuna uses a sampling-based approach to explore more promising configurations efficiently. We tuned the number of layers, units per layer, and learning rate, using logarithmic sampling to cover a wide range of values.

Overall, the parameter ranges were chosen based on common practice and initial experiments, and cross-validation was used to ensure the models generalize well.

E. Results

As observed in *Table 4*, out of the five models evaluated, Logistic Regression achieved the highest level of accuracy and F1-scores at around 0.89. Random Forest and KNN produced levels of accuracy and F1-scores around 0.88, which is just slightly below our best performing model. XGBoost performed just slightly lower, producing F1-scores around 0.87. Finally, Neural Networks produced the lowest performance at an F1-score of around 0.85. Overall, every model performed very well on this dataset, producing very high accuracy and F1-scores, although some models demonstrated slightly better performance than others. See *Table 3* for a summary of the optimal hyperparameters of the trained models.

The main strengths of Logistic Regression are that it is efficient and works well when the data is linear. However, its strength is also its weakness, as it struggles when data is nonlinear. Furthermore, it needs well-preprocessed data as it is sensitive to outliers. Because our data has been preprocessed handling scaling and outliers, this helps prevent two major weaknesses of logistic regression, allowing it to be the best performing model on this dataset.

KNN's key strength is its ability to capture local patterns, especially when the data is well preprocessed and clean. This is because KNN is sensitive to noise when having a lower number of neighbours to look at, such as k=9. Our KNN model performed strongly with our dataset, as it is not an extremely large data set, i.e. fewer computations, along with being well-preprocessed to mitigate noise, playing towards the model's strengths instead of its weaknesses, making it one of the better performing models.

Random Forest's strengths include working well with tabular data, being resistant to noise and outliers, and having less reliance on data preprocessing. However, the reason why our Random Forest model performs well but falls short compared to Logistic Regression could be due to the dataset being well-preprocessed, with outliers already handled. Thus, a well-preprocessed and clean dataset doesn't allow for Random Forest to utilize its strengths fully, making it a well-performing model but not the best.

One of the more advanced models we used on our data is XGBoost. XGBoost's strengths include handling missing data, being highly efficient, and having built-in regularization. Although XGBoost is an advanced model that works well on tabular data, we observe that it is not among the best-performing models in this case. This could be due to the dataset being smaller, simpler, and already well-preprocessed, limiting XGBoost's ability to fully utilize its strengths. Thus, the additional complexity of the model as observed did not provide a significant increase in performance over the simpler models discussed in lecture.

The last model we used on our dataset is Neural Networks, and it is one of the more powerful models we use, which is capable of modeling complex relationships and can produce very high performance. However, this high performance can only be achieved if the model is given enough data to learn from. This is one of its key weaknesses, and likely explains why, despite its capability, it is outperformed by simpler models in this case, due to the relatively small dataset.

F. Future Work

There are possible extensions to this project. As mentioned in the introduction, the data used in this project is a subset of the UCI heart database. Further learning research could be conducted to incorporate more than just the fourteen commonly chosen attributes in the prediction of heart disease, which can also shed light on the statistical weight that different health attributes hold in the prediction and treatment of heart disease. Incorporation of this bigger dataset, and thus, the increase in data dimensionality will also allow us to explore more intricate relationships and optimise the power of models like the artificial neural network.

Another angle is to perform feature importance analysis from models like XGBoost or RandomForest, which would shed light on the weight that certain features hold over heart disease prediction. This insight will inform preventive treatment and early detection of heart disease, contributing to clinical research and heart disease management at a population level.

With a more expansive dataset, other classification models, such as support vector machines or Naive Bayes, could also be explored. This can be done alongside experimenting with techniques such as Synthetic Minority Oversampling Technique (SMOTE), which addresses unbalanced data[1] and improves recall without sacrificing precision, or feature selection using chi-square, ANOVA, and mutual information networks.

Furthermore, a key objective for future development is the transition from static model comparison to a functional predictive tool. Through the introduction of new, unseen clinical observations, the predictive performance of these models can be analyzed to determine the probability of heart disease for each new observation.

[1] Under the assumption that more expansive data means more severe label imbalance introduced

G. References

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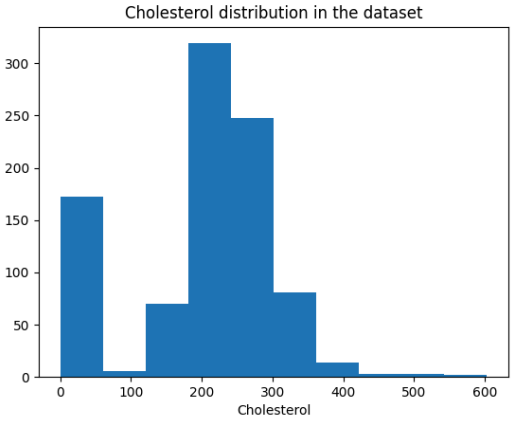
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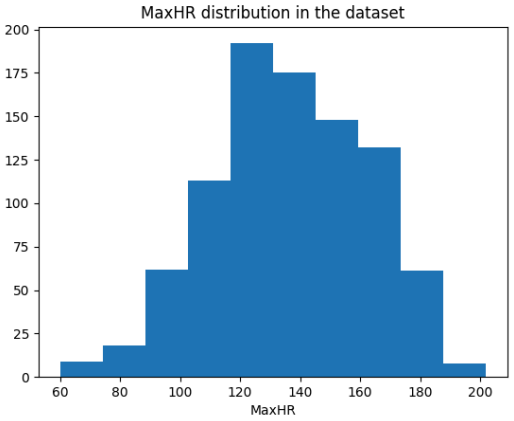
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H. Appendix

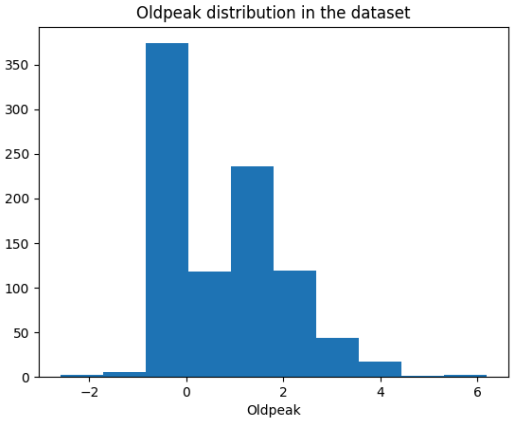
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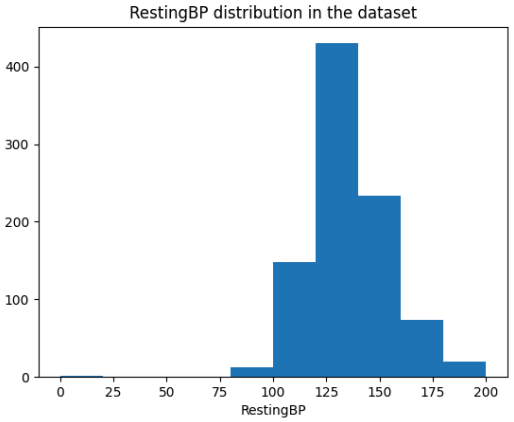
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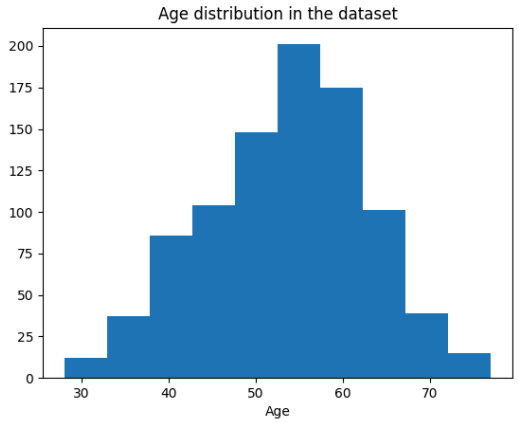


Figure 1. Distribution of numerical features

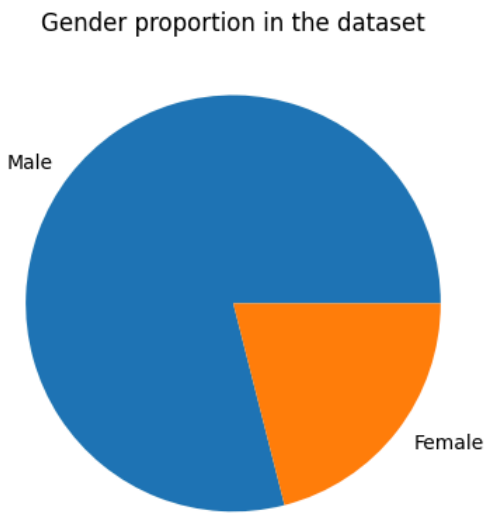


Figure 2. Distribution of gender of participants

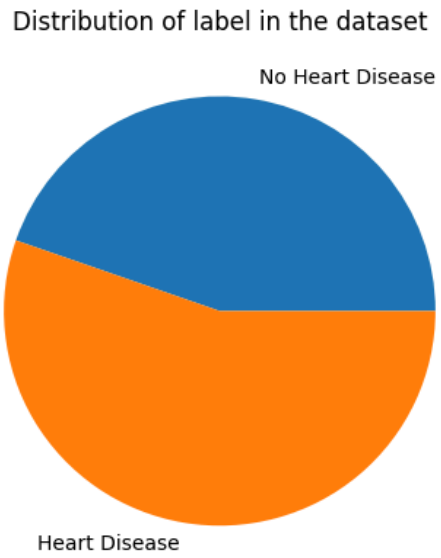
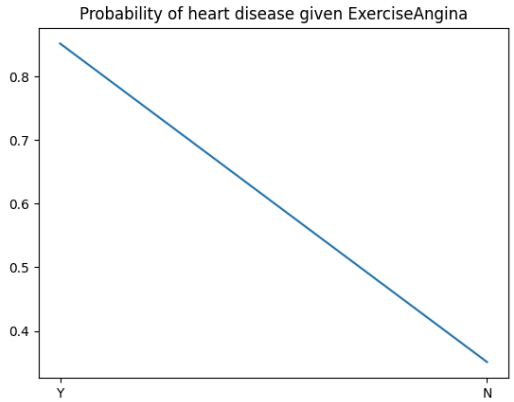
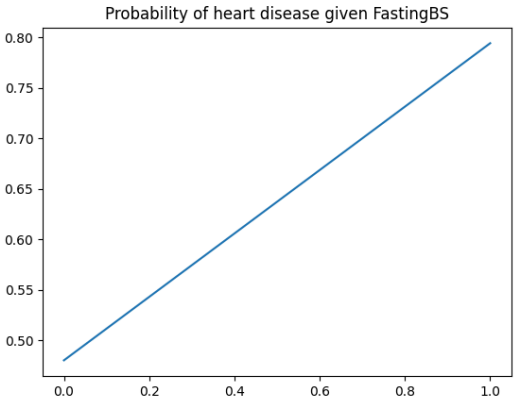


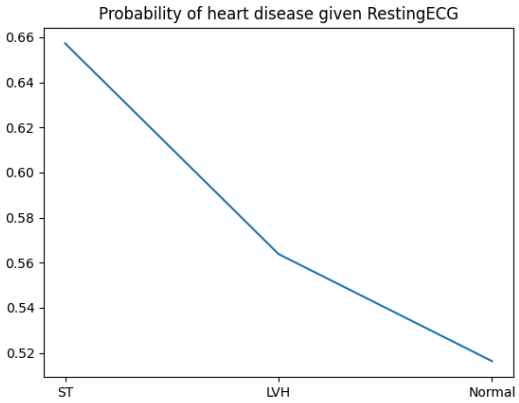
Figure 3. Distribution of label (Heart Disease)



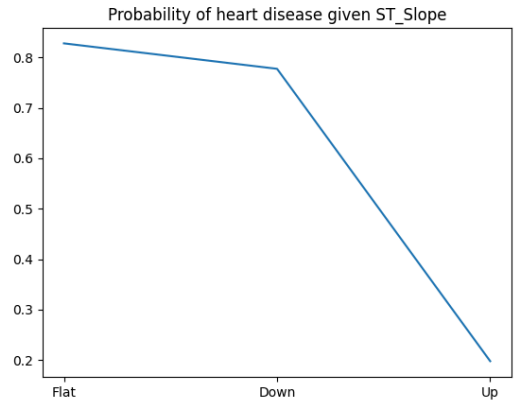
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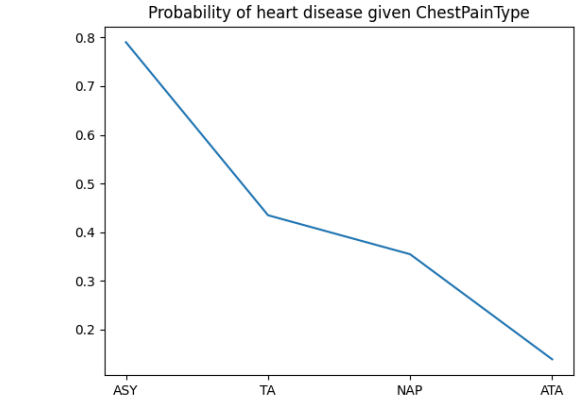
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Figure 4. Probability of heart disease given individual variables

Model	Optimal Hyperparameters
KNN	best params: {'n_neighbors': 9, 'weights': 'uniform'}
Random Forest	Best min samples leaf: {'max_depth': 10, 'min_samples_leaf': 1, 'n_estimators': 100}
Logistic Regression	Model hyperparameters: {'C': np.float64(0.07847599703514611), 'class_weight': None, 'l1_ratio': 0, 'solver': 'liblinear'}
XGBoost	best params: {'colsample_bytree': 1.0, 'gamma': 0.5, 'learning_rate': 0.1, 'max_depth': 7, 'min_child_weight': 3, 'n_estimators': 200, 'reg_alpha': 0.01, 'reg_lambda': 0.1, 'subsample': 0.8}
Neural Network	{'layers': 3, 'units_0': 24, 'units_1': 13, 'units_2': 7, 'lr': 0.0005901744978768212}

Table 3. Optimal Hyperparameters in each model

```
Neural Network Classification Report:
                  precision    recall  f1-score   support

No Heart Disease       0.88      0.80      0.84        82
   Heart Disease       0.85      0.91      0.88       102

        accuracy                           0.86       184
       macro avg       0.87      0.86      0.86       184
    weighted avg       0.87      0.86      0.86       184

KNN Classification Report:
best score: 0.8624072723919701
                  precision    recall  f1-score   support

No Heart Disease       0.84      0.89      0.86        82
   Heart Disease       0.91      0.86      0.88       102

        accuracy                           0.88       184
       macro avg       0.87      0.88      0.87       184
    weighted avg       0.88      0.88      0.88       184
```

```
Random Forest Classification report:
best score: 0.8781410030393972
                  precision    recall  f1-score   support

No Heart Disease       0.85      0.85      0.85        82
   Heart Disease       0.88      0.88      0.88       102

        accuracy                           0.87       184
       macro avg       0.87      0.87      0.87       184
    weighted avg       0.87      0.87      0.87       184


Logistic regression model training report

                  precision    recall  f1-score   support

No Heart Disease       0.88      0.88      0.88        82
   Heart Disease       0.90      0.90      0.90       102

        accuracy                           0.89       184
       macro avg       0.89      0.89      0.89       184
    weighted avg       0.89      0.89      0.89       184


Model hyperparameters: {'C': np.float64(0.07847599703514611), 'class_weight': None,
'solver': 'lbfgs'}
best score: 0.8696911829504093
best params: {'colsample_bytree': 1.0, 'gamma': 0.5, 'learning_rate': 0.1, 'max_depth': 7,
'min_child_weight': 3, 'n_estimators': 200, 'reg_alpha': 0.01, 'reg_lambda': 0.5,
'subsample': 0.8}
best score: 0.8696379262684706
XGBoost report:
                  precision    recall  f1-score   support

No Heart Disease       0.83      0.88      0.85        82
   Heart Disease       0.90      0.85      0.87       102

        accuracy                           0.86       184
       macro avg       0.86      0.87      0.86       184
    weighted avg       0.87      0.86      0.86       184
```

Table 4. Training results with NA handling using KNN imputation